

Linux Server Hardening & Secure Configuration

Objective:

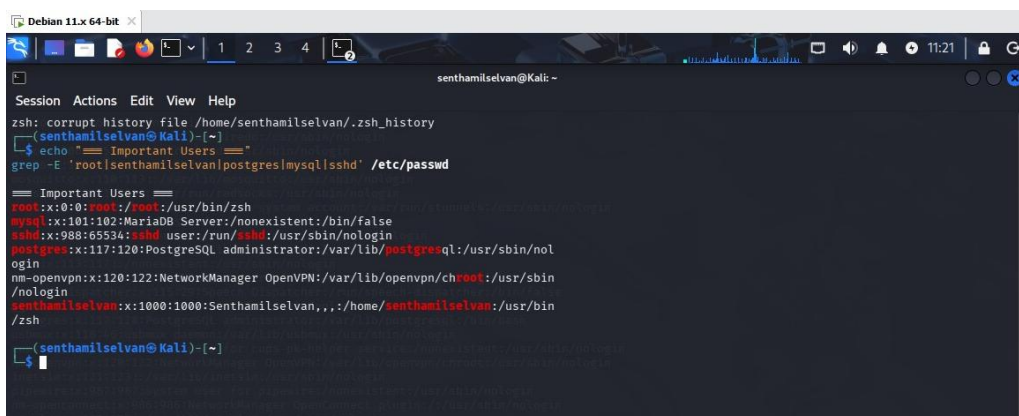
This task was performed to learn and secure the Linux server. This task was performed to remove unwanted users and privilege. This task was done in an isolated lab.

Tools:

Environment: Kali Linux

Unwanted User Removing:

I used “`$ /etc/passwd`” to see available users in my Kali Linux. Then I removed unwanted users and their root privileges. There was some unwanted user have root privilege (postgres), I revoked that. Because of this the system's file are protected from unwanted manipulation.



```
Debian 11.x 64-bit x
Session Actions Edit View Help
zsh: corrupt history file /home/senthamilselvan/.zsh_history
(senthamilselvan@Kali)-[~]
$ echo "=== Important Users ==="
grep -E 'root|senthamilselvan|postgres|mysql|sshd' /etc/passwd

=== Important Users ===
root:x:0:0:root:/root:/usr/bin/zsh
mysql:x:101:102:MariaDB Server:/nonexistent:/bin/false
sshd:x:988:65534:sshd user:/run/ssh:/usr/sbin/nologin
postgres:x:117:120:PostgreSQL administrator:/var/lib/postgresql:/usr/sbin/nologin
nm-openvpn:x:120:122:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
senthamilselvan:x:1000:1000:Senthamilselvan,,,:/home/senthamilselvan:/usr/bin/zsh

(senthamilselvan@Kali)-[~]
```

Risk:

Moderate

Recommendation:

- Remove unknown users.
- Revoke unwanted privileges.

Unwanted Services Disabling:

I used “\$ sudo systemctl list-units –type=service –state=running” to how many services are running on the Linux. In the victim device there was lot of services running. But some of them were unwanted to the device usage. Those are “docker.service”, “containerd.service” and also the system doesn’t need any physical connection for network. So, “ModemManager.service” also unrelated to the system. So, stopped and disable those services.

```
(senthamilselvan@Kali)-[~]
$ sudo systemctl stop docker
Stopping 'docker.service', but its triggering units are still active:
docker.socket

(senthamilselvan@Kali)-[~]
$ sudo systemctl disable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable docker
Removed '/etc/systemd/system/multi-user.target.wants/docker.service'.
Disabling 'docker.service', but its triggering units are still active:
docker.socket

(senthamilselvan@Kali)-[~]
$ sudo systemctl stop containerd

(senthamilselvan@Kali)-[~]
$ sudo systemctl disable containerd

(senthamilselvan@Kali)-[~]
$ sudo systemctl stop ModemManager

(senthamilselvan@Kali)-[~]
$ sudo systemctl disable ModemManager
Removed '/etc/systemd/system/dbus-org.freedesktop.ModemManager1.service'.
Removed '/etc/systemd/system/multi-user.target.wants/ModemManager.service'.

(senthamilselvan@Kali)-[~]
$
```

Risk:

High

Recommendations:

- Always stop the services after it’s work.
- Check daily once for the running services.

SSH Hardening:

The SSH in the victim device was allowing users to give password for the connection. It made the device to vulnerable for brute force attacks. So, I changed it into encrypted key pairs. Now, the SSH doesn’t ask passwords for connection instead of that it ask for key

```
(senthamilselvan@Kali)-[~]
$ sudo sshd -T | grep -E "permitrootlogin|passwordauthentication"

permitrootlogin no
passwordauthentication yes
```

pair.

```
(senthamilselvan@Kali)-[~]
$ sudo sshd -T | grep -E "passwordauthentication|permitrootlogin"
permitrootlogin no
passwordauthentication no
```

Risk:

Low

Recommendations:

- Always use key-pairs for high security.
- Use passphrase also for high security.

Configuring Firewall:

In victim server there is no firewall installed. So installed firewall using “\$ sudo apt install ufw”.

```
(senthamilselvan@Kali)-[~]
$ sudo apt install ufw
Installing:
 ufw

Suggested packages:
 rsyslog

Summary:
 Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1602
 Download size: 169 kB
 Space needed: 880 kB / 1,655 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 ufw all 0.36.2-9 [169 kB]
Fetched 169 kB in 1s (212 kB/s)
Preconfiguring packages ...
Selecting previously unselected package ufw.
(Reading database ... 422779 files and directories currently installed.)
Preparing to unpack .../archives/ufw_0.36.2-9_all.deb ...
Unpacking ufw (0.36.2-9) ...
Setting up ufw (0.36.2-9) ...
Creating config file /etc/ufw/before.rules with new version
Creating config file /etc/ufw/before6.rules with new version
Creating config file /etc/ufw/after.rules with new version
Creating config file /etc/ufw/after6.rules with new version
update-rc.d: We have no instructions for the ufw init script.
update-rc.d: It looks like a non-network service, we enable it.
Created symlink '/etc/systemd/system/multi-user.target.wants/ufw.service' -> '/usr/lib/systemd/system/ufw.service'.
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for man-db (2.13.1-1) ...
```

After that I allowed only SSH (port - 22). Then I blocked other all port's traffic.

```
(senthamilselvan@Kali)-[~]
$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)

(senthamilselvan@Kali)-[~]
$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)

(senthamilselvan@Kali)-[~]
$ sudo ufw allow ssh
Rules updated
Rules updated (v6)

(senthamilselvan@Kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup
```

Firewall Status:

```
(senthamilselvan@Kali)-[~]
$ sudo ufw status verbose
[sudo] password for senthamilselvan:
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
```

Configuring Auto Updates:

I enabled the auto package updates on victim machine. Because of this the machine can do updates without any user interaction. So, the vulnerabilities might come lower then usual.

```
(senthamilselvan@Kali)-[~]  
$ sudo dpkg-reconfigure --priority=low unattended-upgrades  
Creating config file /etc/apt/apt.conf.d/20auto-upgrades with new version
```

File Access:

I carefully gave read, write, and execute permissions for user. Because of the file manipulation are decreased. The important files like /etc/passwd, /etc/shadow, /etc/group are become secured.

```
(senthamilselvan@Kali)-[~]  
$ sudo chmod 644 /etc/passwd  
sudo chmod 600 /etc/shadow  
sudo chmod 644 /etc/group  
sudo chmod 600 /etc/gshadow  
sudo chmod 440 /etc/sudoers
```